

Abhay Tomar

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Summary

I am a passionate and results-driven Software Engineer with a solid foundation in both software development and embedded systems programming. Proficient in C, C++, PHP, JavaScript and ReactJS with hands-on expertise in microcontroller programming (ESP32, STM32), embedded C, RTOS, and communication protocols (UART, SPI, I2C). Skilled in full-stack development, database integration, and IoT solutions. Passionate about building scalable, reliable software for both digital platforms and hardware-based systems.

Work Experience

Software Engineer, SVURG Digital System Solutions – Jasola, Delhi Sep 2024 – Present

- Designed, developed and maintain responsive web interfaces using HTML, CSS, JavaScript, Tailwind, and PHP.
- Developed and maintained firmware for microcontrollers (ESP32, STM32) using C/C++, supporting interfaces like UART, SPI, I2C, and GPIO.
- Integrated hardware peripherals such as TFT touch displays, TMC2209 stepper drivers, AS5600 sensors, SD cards, USB (CH376msc), and rotary encoders.
- Debugged, tested, and optimized code for real-time responsiveness and memory efficiency.
- Built communication bridges between web applications and embedded systems via Bluetooth, serial (UART), and HTTP protocols.

Web Developer, Masterguard Protective Services Pvt. Ltd. – Mayur Vihar, Delhi June 2022 - Aug 2024

- Maintained and updated website content, including text, images, links, and downloadable files.
- Ensured pages were functioning correctly across different browsers and devices.
- Monitored website performance, fixed broken links, and ensured smooth user navigation.
- Performed regular backups of the website and ensured all updates were safely stored.

Education

IGNOU University, Bachelor in Computer Applications(BCA) July 2021 – June 2024

- GPA: 7/10
- **Coursework:** Learned Web technologies, data structures, DBMS, operating systems, and software engineering.

Technologies

Languages: C/C++, Embedded C, HTML/CSS, JavaScript, NodeJS, PHP

Frameworks and Tools: Tailwind, ReactJS, WordPress

Protocols: UART, SPI, I2C, WIFI, BT, BLE

IDE: Arduino, VS Code

Version Control and Database: Git and Github, MySQL

Projects

Currently working on the development and maintenance of the TDR Foundation website.(truedigitalradio.org) and Svurg website(svurg.com)

- Developed and maintained product and service pages.
- Performed regular content updates and feature enhancements.
- Integrated third-party tools such as payment gateways and email marketing services.

- Optimized performance and ensured site security and uptime.
- Handled data backup, recovery processes, and ensured legal compliance.

Worked on Masterguard company's website(masterguardindia.com)

- Developed and maintained the official website for Masterguard..
- Improved features and performance of the website. Mobile Responsiveness and Performance.

Stepper Motor Controller (ESP32)

- Developed an embedded system to control a stepper motor using ESP32
- Implemented a TFT touchscreen UI for motor control with real-time feedback.
- Designed hierarchical menu navigation using a rotary encoder and button controls.

CNC Controller(STM32H747)

- Project Objective: Develop a high-performance CNC controller using the STM32H747 microcontroller, leveraging its dual-core architecture for real-time multitasking. The M7 core handles motion control, while the M4core manages peripheral communication, ensuring precision, efficiency, and scalability while enhancing system responsiveness and integration with CNC hardware and software.
- Developed an embedded system(Magnetic meter) to use this as a feedback system for CNC Machine for checking X, Y and Z coordinates using AS5600 magnetic sensor and MSPMOL1306 controller.